Name: _____ **Class:** _____

JURONG PIONEER JUNIOR COLLEGE

JC2 Preliminary Examination 2026

PHYSICS **9478/03**
Higher 2 **14 September 2026**

Paper 3 Longer Structured Questions **2 hours**

You must answer on the question paper.
No additional materials are needed.

INSTRUCTIONS

- Section A: answer **all** questions.
- Section B: answer **one** question.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name and class at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

For Examiner's Use	
1	/ 7
2	/ 9
3	/ 8
4	/ 7
5	/ 8
6	/ 9
7	/ 7
8	/ 20
9	/ 20
Total	/ 75

This document has **28** pages. Any blank pages are indicated.

Data

speed of light in free space	$c = 3.00 \times 10^{8}$ m s^{-1}
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9} \text{ m F}^{-1}\right)$
elementary charge	$e = 1.60 \times 10^{-19}$ C
Planck constant	$h = 6.63 \times 10^{-34}$ J s
unified atomic mass constant	$u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ J K^{-1} mol^{-1}
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	$k = 1.38 \times 10^{-23}$ J K^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
acceleration of free fall	$g = 9.81$ m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
work done on/by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T / \text{K} = T / {}^{\circ}\text{C} + 273.15$

pressure of an ideal gas $p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$

mean translational kinetic energy of an ideal gas molecule $E = \frac{3}{2}kT$

displacement of particle in s.h.m. $x = x_0 \sin \omega t$

velocity of particle in s.h.m. $v = v_0 \cos \omega t = \pm\omega\sqrt{x_0^2 - x^2}$

electric current $I = nAvq$

resistors in series $R = R_1 + R_2 + \ldots$

resistors in parallel $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

capacitors in series $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$

capacitors in parallel $C = C_1 + C_2 + \ldots$

energy in a capacitor $U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$

charging a capacitor $Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$

discharging a capacitor $Q = Q_0 e^{-\frac{t}{\tau}}$

RC time constant $\tau = RC$

electric potential $V = \frac{Q}{4\pi\varepsilon_0 r}$

alternating current/voltage $x = x_0 \sin \omega t$

magnetic flux density due to a long straight wire $B = \frac{\mu_0 I}{2\pi d}$

magnetic flux density due to a flat circular coil $B = \frac{\mu_0 NI}{2r}$

magnetic flux density due to a long solenoid $B = \mu_0 nI$

energy states for quantum particle in a box $E_n = \frac{h^2}{8mL^2}n^2$

radioactive decay $x = x_0 e^{-\lambda t}$

radioactive decay constant $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** questions in the spaces provided

1 **(a)** In an experiment, two objects A and B are released from the side of a building, as shown in Fig. 1.1.

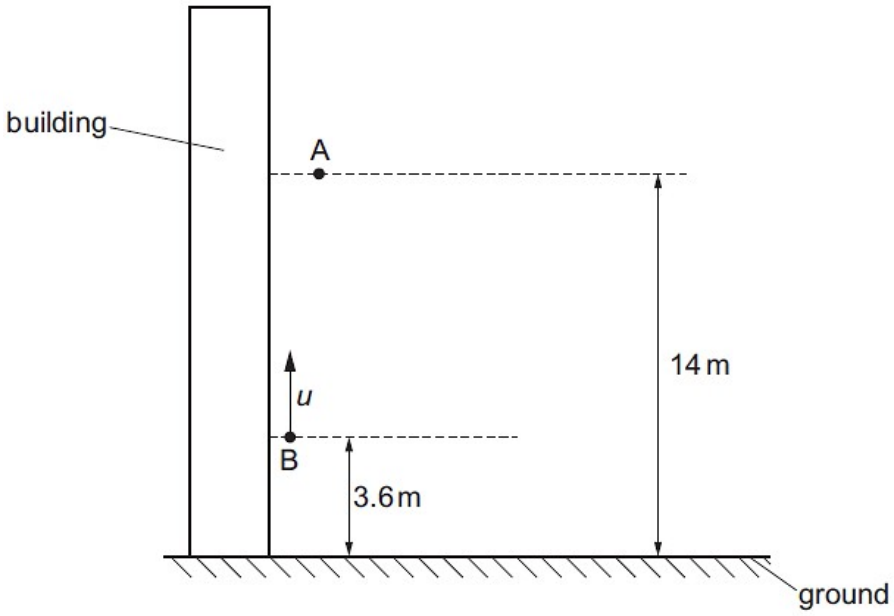


Fig. 1.1 (not to scale)

Object A is released from rest at a height of 14 m above the horizontal ground.
Object B is released with an initial upwards vertical velocity u at a height of 3.6 m above the ground.
Both objects take the same time t to reach the ground and they do not collide with each other.

Assume that air resistance is negligible.

Calculate u.

u = ... m s^{-1} [3]

(b) In a second experiment, object B is released from the same height and given the same initial speed as in **(a)** but at a release angle θ to the vertical, as shown in Fig. 1.2.

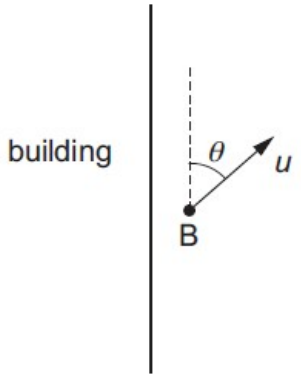


Fig. 1.2

(i) State and explain whether the time taken for object B to reach the ground is less than, the same as or greater than the time t in **(a)**.

……

……

…… [2]

(ii) State and explain the effect of the change in release angle on the speed at which B reaches the ground.

……

……

…… [2]

2 A small ball is dropped from rest and falls vertically downwards. The ball collides with the ground and bounces back vertically upwards. Fig. 2.1 shows the ball just before and just after hitting the ground.

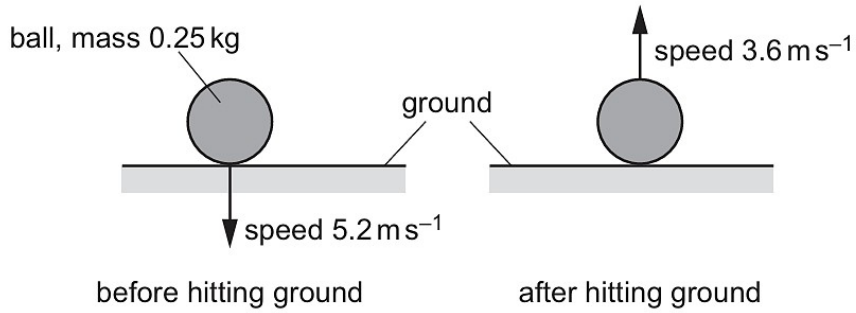


Fig. 2.1

The ball has mass 0.25 kg and is in contact with the ground for a time of 0.18 s. Just before the ball hits the ground, it has speed 5.2 m s^{-1}. Just after it leaves the ground, it has speed 3.6 m s^{-1}.

Air resistance acting on the ball is negligible.

(a) State and explain whether the collision is elastic or inelastic.

….....

….....

…..... [1]

(b) (i) Determine the change in momentum of the ball during the collision with the ground.

magnitude of change in momentum = ... kg m s^{-1}

direction of change = …… [2]

(ii) Assume that the force on the ball during the collision with the ground is constant.

1. Calculate the force on the ground during the collision.

force = ... N [3]

2. The ball hits the ground at time *t*.

On Fig. 2.2, sketch the variation with time of the resultant force on the ball.

Label the resultant force axis with numerical values.

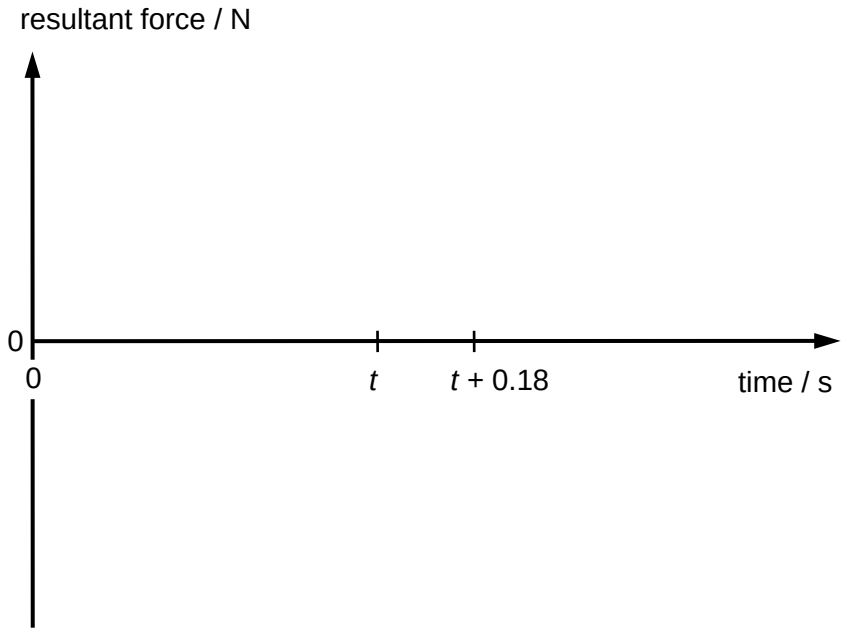


Fig. 2.2

[3]

3 An uniform metal bar AB makes an angle of 30° with the horizontal ground, as shown in Fig. 3.1.

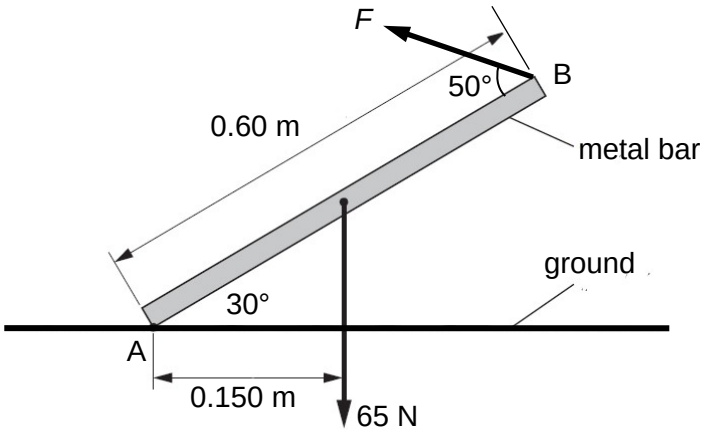


Fig. 3.1

The bar is hinged at A and is supported by a rope at B. The force *F* produced by the rope at B acts at an angle of 50° to the bar.

(a) The bar has a length of 0.60 m and a weight of 65 N.

Calculate the magnitude of force *F*.

F = ... N [3]

(b) State and explain how your answer to **(a)** would change if the force *F* is applied at a smaller angle to the bar.

.....

……..……

…… [2]

(c) A force *R* acts on the bar at A.

Calculate the magnitude of *R*.

R = ... N [3]

4 A loudspeaker is connected to a signal generator producing a sinusoidal output of frequency 650 Hz as shown in Fig. 4.1. The sound waves generated pass through points X and Y, directly in front of the loudspeaker.

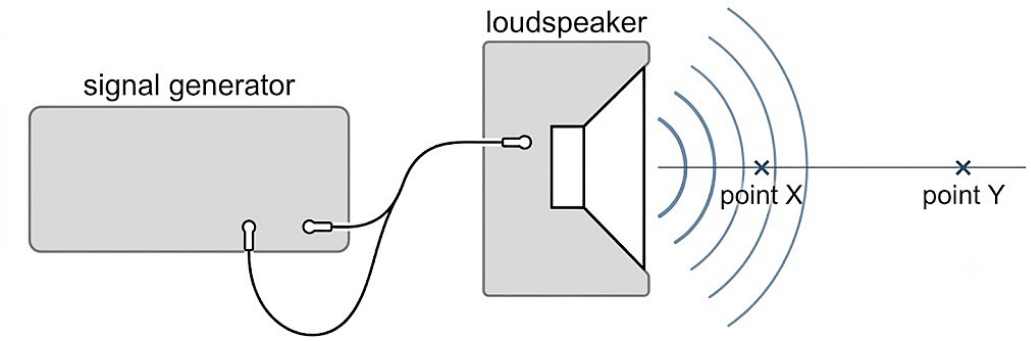


Fig. 4.1

(a) The waves emitted by the loudspeaker are longitudinal.

Explain the meaning of polarisation of wave and hence why longitudinal waves cannot be polarised.

.....

.....

.....

..... [2]

(b) Fig 4.2 shows the variation with time t of Δp, the excess pressure above the atmospheric pressure at point X as the sound passes through it.

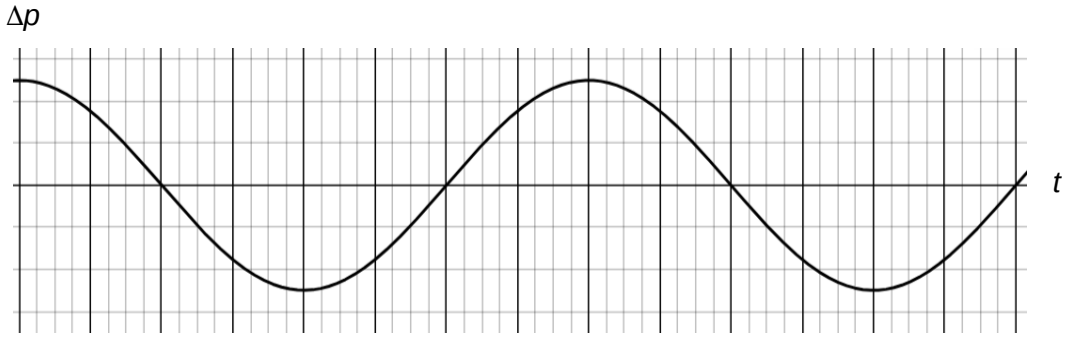


Fig. 4.2

(i) On Fig. 4.2, sketch a graph showing the variation of the displacements of the air molecules at X with time, taking right as the positive direction.

[2]

(ii) The phase difference between the waves at X and Y is 3.50 rad. X and Y are separated by 0.300 m.

Determine the speed of the sound wave.

speed = …… m s^{-1} [2]

(iii) Suggest why the amplitude does not decrease inversely with distance from the loudspeaker.

.....

…..... [1]

5 **(a)** A battery has an e.m.f. of 4.5 V and is connected to a 48 Ω resistor. The potential difference across the resistor is found to be 4.0 V.

(i) Calculate the internal resistance of the battery.

internal resistance = ... Ω [2]

(ii) Determine the energy dissipated in the 48 Ω resistor in a time of 2 minutes.

energy = ... J [1]

(b) Fig. 5.1 shows the I-V characteristics of an electrical component C when the potential difference across C is varied from 0 V to 10 V.

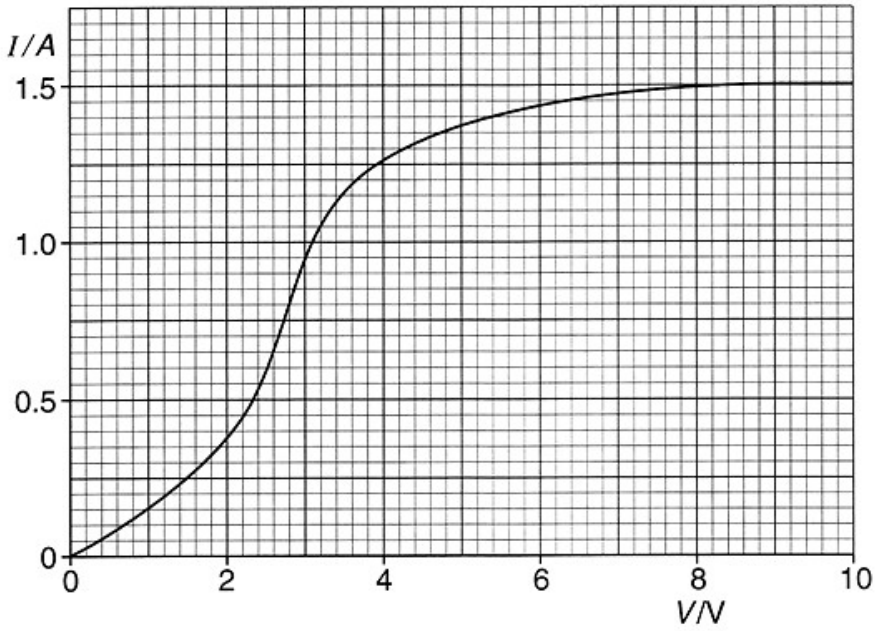


Fig. 5.1

(i) By drawing an appropriate line in Fig. 5.1, show that the minimum value of the resistance of component C is approximately 3.0 Ω.

[2]

(ii) Explain how the answer to **(a)(ii)** will change when component C is added to the 48 Ω resistor in parallel.

.....

.....

.....

..... [1]

(c) The resistance R of a uniform metal wire is measured for different lengths l of the wire. The variation with l of R is shown in Fig. 5.2.

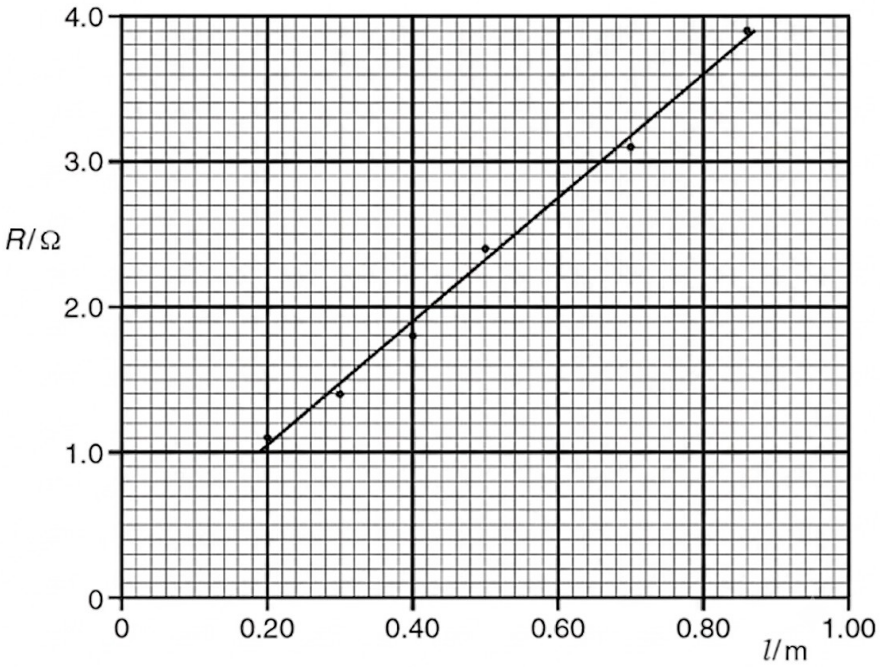


Fig. 5.2

(i) Determine the gradient of the line in Fig. 5.2.

gradient = ... [1]

(ii) The cross-sectional area of the wire is 0.12 mm^2.

Use your answer in **(i)** to determine the resistivity of the metal of the wire.

resistivity = ... Ω m [1]

6 Fig. 6.1 shows a rotating coil of a simple electrical generator. A rectangular coil of wire is rotated in a uniform magnetic field of magnetic flux density 3.0×10^{-2} T. The axis of rotation is perpendicular to the direction of the magnetic field. The axis of the coil makes an angle θ to the direction of the magnetic field.

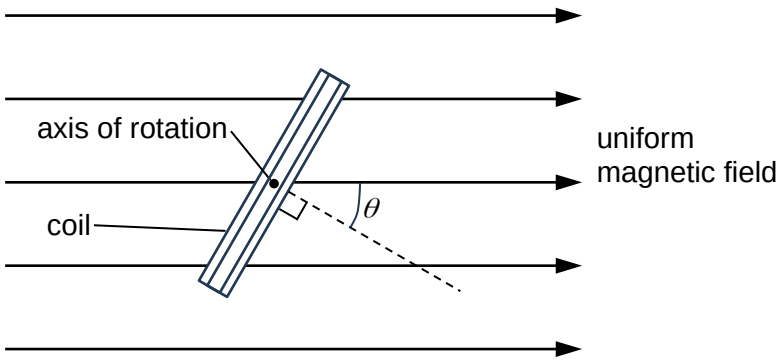


Fig. 6.1

(a) The coil has 250 turns and a cross-sectional area of 2.4 cm^2.

Show that the magnetic flux linkage for the coil is 1.8×10^{-3} Wb when $\theta = 0$.

[1]

(b) The coil is rotated at a constant rate of 2 revolutions per second.

(i) Calculate the magnitude of the average electromotive force (e.m.f.) induced in the time taken for the coil to rotate from $\theta = 0$ to $\theta = 45^\circ$.

e.m.f. = ... V [2]

(ii) Fig. 6.2 show the variation with θ of the induced e.m.f. over one cycle of rotation of the coil.

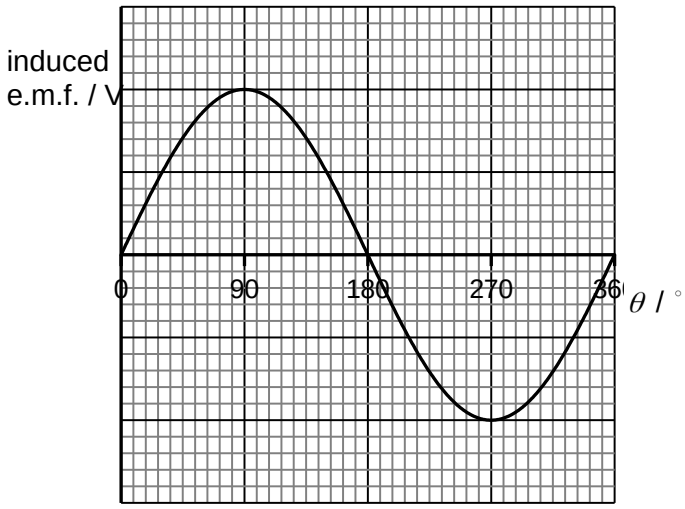


Fig. 6.2

Explain why the induced e.m.f. is sinusoidal.

……

……

……

……

…….. [2]

(c) A small solenoid P is placed inside a larger solenoid Q, as shown in Fig. 6.3.

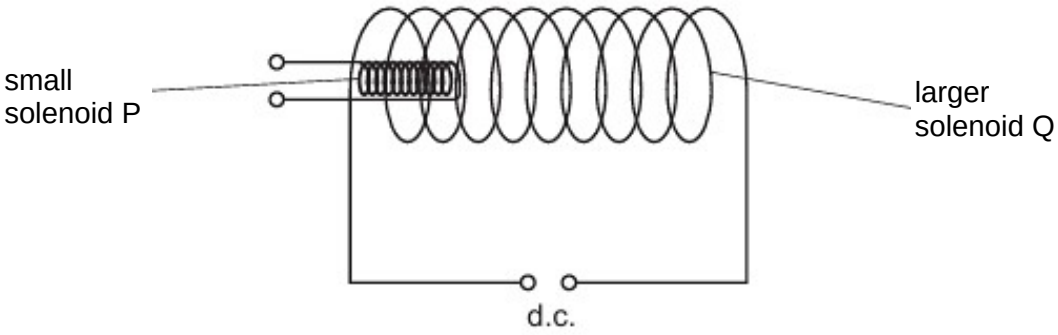


Fig. 6.3

Solenoid Q is attached to a d.c. power supply to create a magnetic field.

(i) State Lenz's law of electromagnetic induction.

……

……

…….. [1]

(ii) The terminals of P are connected together. P is then removed from inside Q.

With reference to magnetic fields, explain why a force is needed to remove P.

……

……

……

……

……

…….. [3]

7 **(a)** In quantum mechanics, the state of a particle is described by a wavefunction ψ.

State the physical significance of $|\psi|^2$.

.....

..... [1]

(b) An electron is confined in a one-dimensional infinite square well of width L. The wavefunctions of the three lowest energy levels are shown in Fig. 7.1.

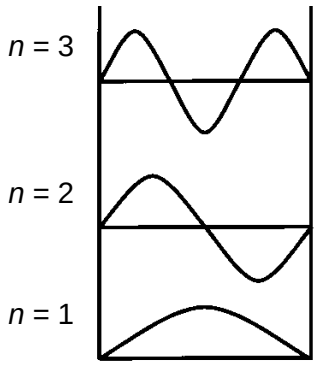


Fig. 7.1

(i) Explain how superposition leads to these wavefunctions shown in Fig. 7.1.

.....

.....

.....

..... [2]

(ii) Determine the probability of finding the electron in the interval x where $\frac{L}{3} \leq x \leq \frac{L}{2}$ when the electron is at the energy level $n = 3$.

probability = ... [2]

(c) An electron is confined in an infinite square well of width $L = 2.0 \times 10^{-10}$ m.

Calculate the energy required to excite the electron from the ground state to the first excited state.

energy = ... J [2]

Section B

Answer **one** question in this section in the spaces provided.

8 (a) (i) State what is meant by a *field of force*.

……

……

…… [1]

(ii) Define *electric field strength*.

……

…… [1]

(iii) Suggest why, when defining electric field strength, the test particle must be stationary.

……

…… [1]

(b) (i) State the relation between electric field strength E and potential V.

…… [1]

(ii) Two charged metal spheres A and B, of diameters 18 cm and 12 cm respectively, are isolated in space, as shown in Fig. 8.1.

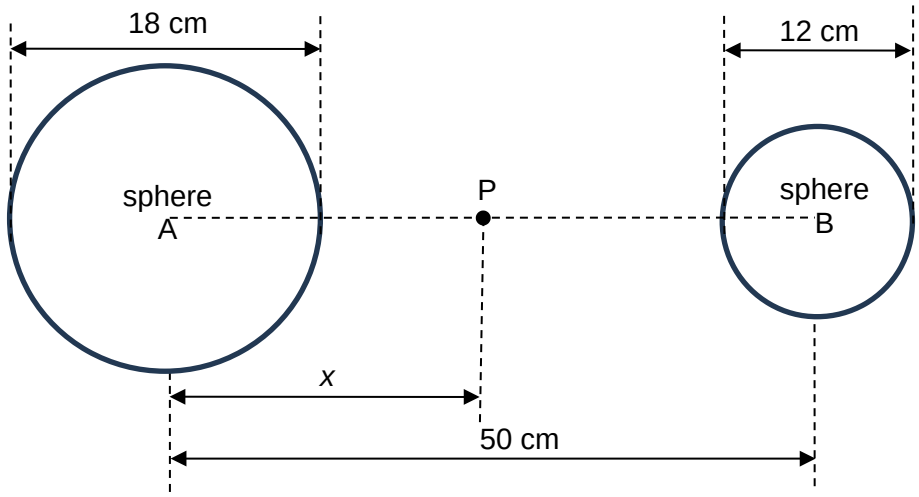


Fig. 8.1

The centres of the spheres are separated by a distance of 50 cm. Point P is at a distance x from the centre of sphere A along the line joining the centres of the two spheres.

The variation with x of the electric potential V at P is shown in Fig. 8.2.

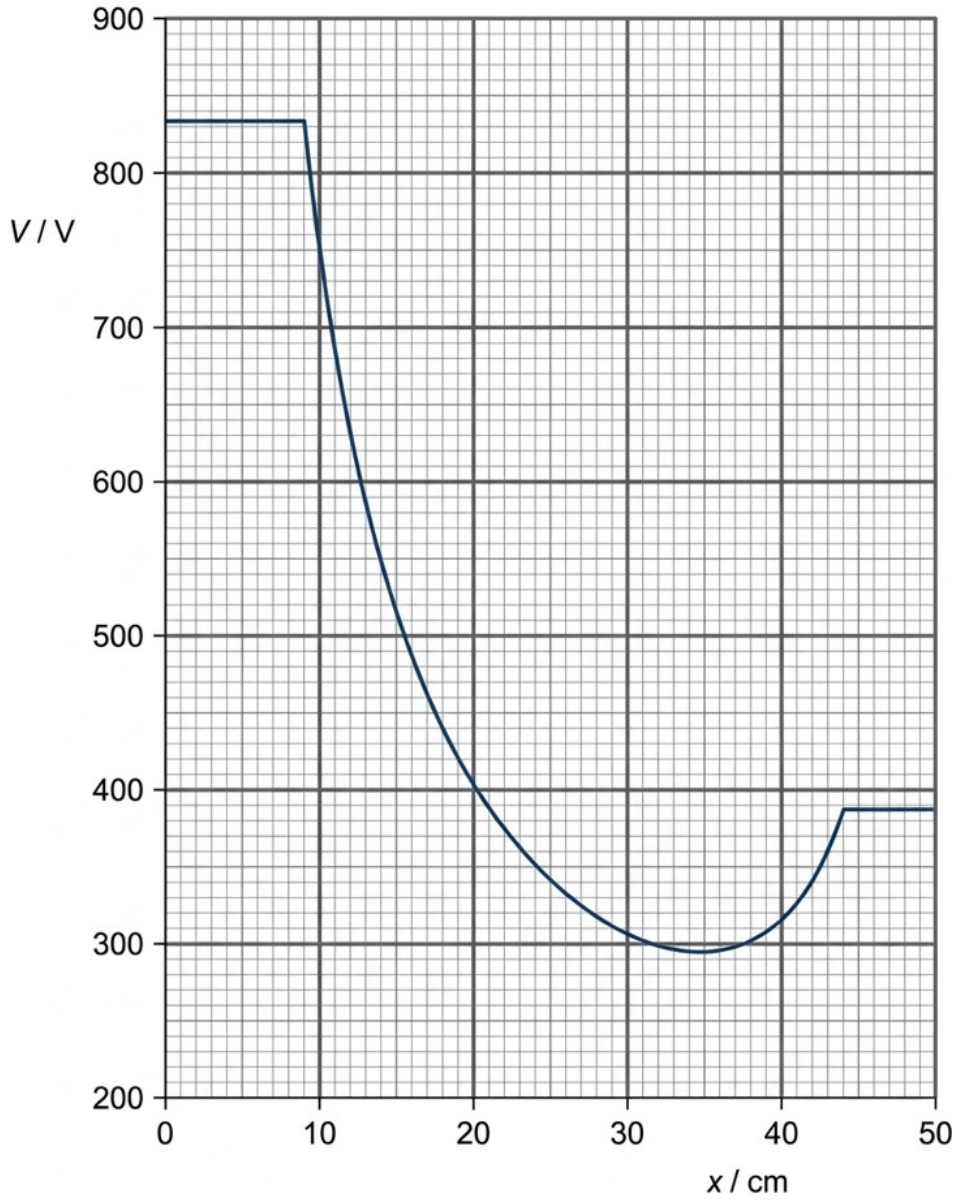


Fig. 8.2

1. State and explain the direction of the electric field at the point P, where x = 25.0 cm.

……

……

…… [2]

2. Use Fig. 8.2 to determine the force on an electron placed at point P, where x = 35.0 cm.

force = …… N [3]

3. By making reference to electric fields, explain why the potential is constant for distances between x = 0 and x = 9.0 cm.

……

……

…… [2]

(c) An electron, initially at rest a long distance from the spheres in **(b)**, approaches the spheres and passes between the two spheres.

Calculate the minimum speed of the electron as it crosses the line joining the centres of the two spheres.

speed = …… m s^{-1} [3]

(d) A capacitor is connected in a circuit used for a flash camera as shown in Fig. 8.3.

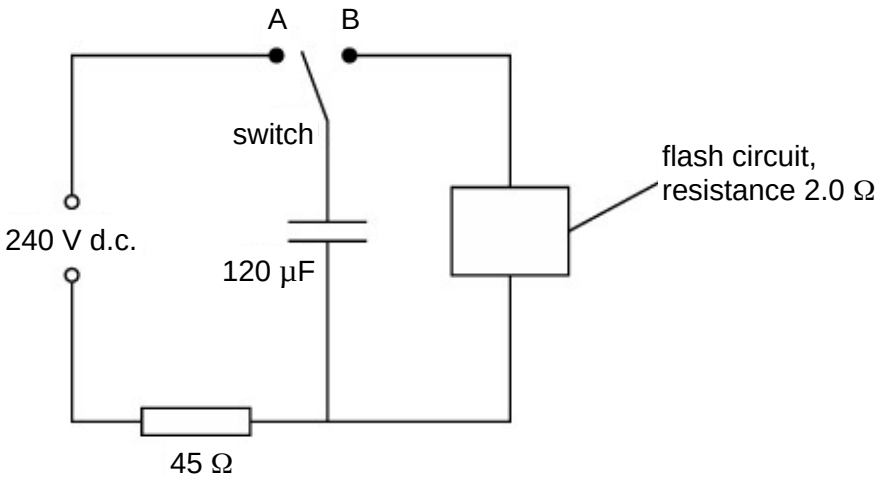


Fig. 8.3

(i) When the switch is connected to terminal A, the capacitor charges.

The switch automatically connects to terminal B when the charge in the capacitor reaches 95% of its maximum possible charge at this voltage.

Calculate the time taken for the capacitor to reach 95% of its maximum charge.

time taken = ……. s [3]

(ii) The capacitor now discharges through the flash circuit.

Calculate the maximum amount of energy that the capacitor can deliver to the flash circuit.

energy = …… J [3]

9 **(a)** An unstable nucleus of nucleon number (mass number) A undergoes α-decay as illustrated in Fig. 9.1.

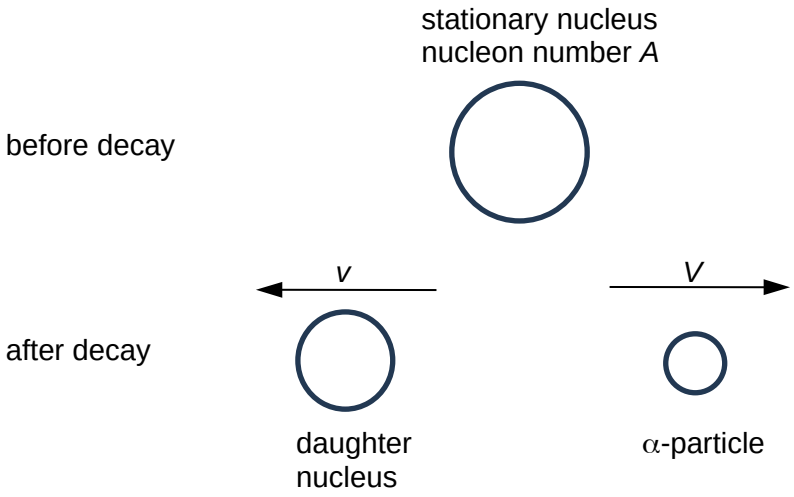


Fig. 9.1

The nucleus is stationary before the decay.

After the decay, the initial speed of the α-particle is V and that of the daughter nucleus is v.

(i) State an equation, in terms of A, v and V, to represent conservation of linear momentum for this decay.

……

…… [1]

(ii) Show that the ratio

$$\frac{\text{initial kinetic energy of } \alpha\text{-particle}}{\text{initial kinetic energy of daughter nucleus}}$$

is equal to $\left(\frac{1}{4}A - 1\right)$.

[3]

(b) Data for the α-decay of bismuth-212 ($^{212}_{83}Bi$) to form thallium-208 ($^{208}_{81}Tl$) are given in Fig. 9.2.

nucleus	mass of nucleus / u
bismuth-212	211.9459
thallium-208	207.9374
helium-4	4.0015

Fig. 9.2

(i) Use the data of Fig. 9.2 to calculate, to two places of decimals, the energy released during the decay.

energy = …… MeV [3]

(ii) Use your answer in **(i)** to show that, based on the expression in **(a)(ii)**, the energy of the α-particle is 6.42 MeV.

[2]

(c) In practice, the α-particle is found to have an energy of 6.10 MeV, rather than 6.42 MeV, as calculated in **(b)(ii)**.

Suggest

(i) an explanation for the difference in energy,

……

…… [1]

(ii) why it is likely that the thallium nucleus and the α-particle do not move off in opposite directions.

……

……

……

……

…… [3]

(d) Some data for the half-lives and decay constants of bismuth-212 and thallium-208 are given in Fig. 9.3.

nucleus	half-life / s	decay constant / s^{-1}
bismuth-212	……	
thallium-208	190	3.7×10^{-3}

Fig. 9.3

(i) Define

1. half-life,

……

…… [1]

2. decay constant.

……

…… [1]

(ii) Show that decay constant λ and half-life $t_{\frac{1}{2}}$ of a radioactive substance are related by $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$.

[2]

(iii) Complete Fig. 9.3 by calculating the half-life of bismuth-212.

[1]

(iv) Initially, a radioactive source contains *N* nuclei of bismuth-212.

After two hours, it is found that the number of bismuth-212 nuclei has reduced to approximately $\frac{1}{4}N$. However, although bismuth-212 decays to form thallium-208, the number of thallium nuclei is much less than $\frac{3}{4}N$.

Suggest an explanation for these observations.

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……

……

…… [2]

End of paper